

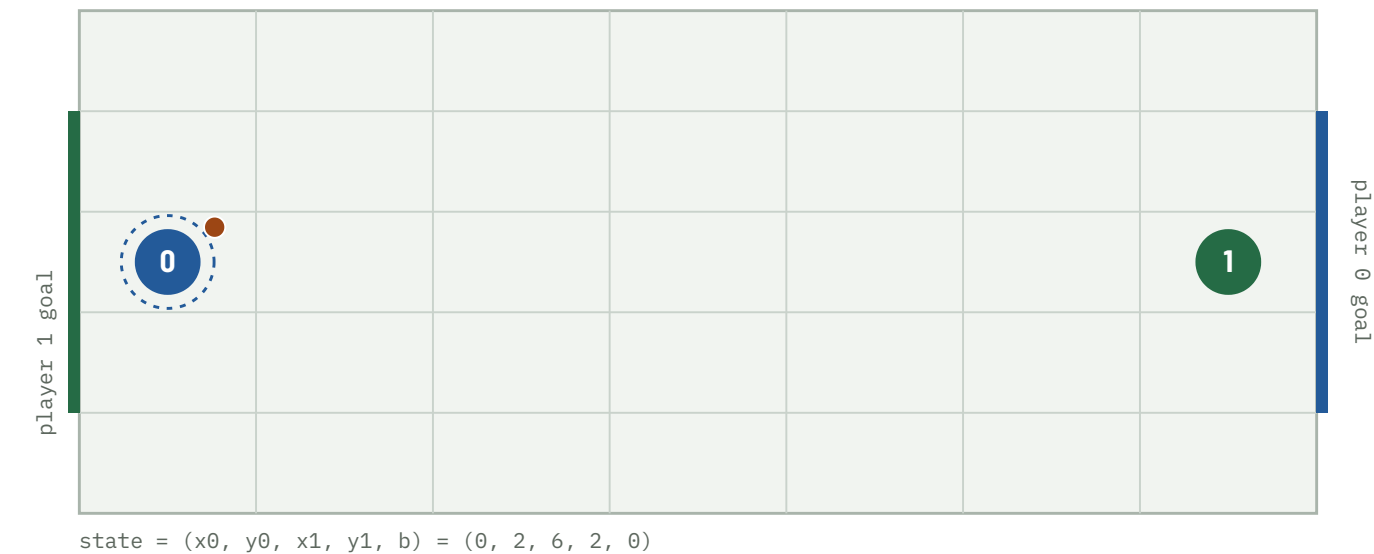
# Pure-First Nash Q for Soccer

Where a two-player soccer Markov game does — and does not — need mixed strategies, and how much matrix-game work that saves.

FOUR QUESTIONS

**RQ1 Existence** — under what transition structures does a pure equilibrium exist at every state? **RQ2 Efficiency** — how much work does checking for a pure saddle first eliminate? **RQ3 Mechanism** — which spatial configurations force mixing? **RQ4 Robustness** — how do discounting and tolerance affect the split?

repo · soccer-markov-nash | 150+ tests | tables from experiments/\*.csv | scope in docs/assumptions.md



**The kickoff.** 7×5 grid, 2380 non-terminal states. Player 0 (blue) carries the ball and attacks the right edge; player 1 (green) the left. Actions U D L R are chosen simultaneously; reward is +1 / -1 / 0, zero-sum.

## ABSTRACT

- 1** For the implemented deterministic A10 model, every one of the 2380 converged stage games has a **pure saddle**. Playing a pure saddle action everywhere is then a stationary pure-strategy equilibrium, found in 9 sweeps with no linear program.

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- 2** A **coin-flip tie-break does not change this**; Littman's random *move order* does — but only when the goal mouth is more than one cell wide, and even then on ~4% of states (this configuration).

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- 3** Those mixed states are **geometrically localised**: a decision tree predicts them from the state with precision 0.96, and 68 of 94 have a 2×2 matching-pennies support.

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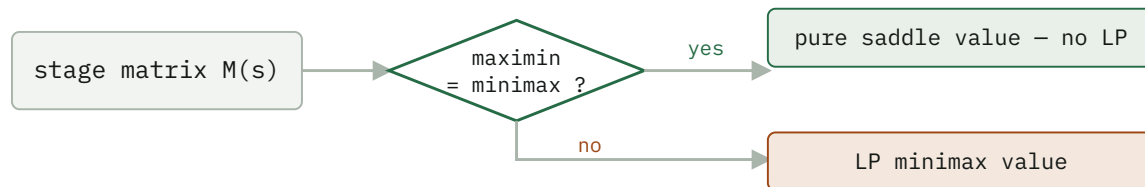
- 4** The **pure-first hybrid solver** matches the all-LP value function to 4e-16 while doing ~25× less matrix-game work; mirror symmetry halves it again.

## Method

The game is zero-sum, so at every state the stage game is a payoff matrix  $M(s)[a_0, a_1] = E[R + \gamma V(s')]$  and the Shapley (1953) fixed point is the minimax value:

$$\begin{aligned} V(s) &= \text{val}(M(s)) = \max_n \min_{k \in \mathcal{K}} (p^T M)[a_1] \\ &= \min_p \max_{a_0} (Mq)[a_0] \end{aligned}$$

The  $Q = R + \beta \text{Nash}(Q')$  update is this operator. Three stage backups: **pure** (the maximin security value — a fast lower bound, not a Nash value when it is below the minimax), **mixed** (the LP minimax value, via `scipy HiGHS`), and **hybrid** (the pure saddle value where maximin = minimax, the LP elsewhere — exact everywhere). Support enumeration (`support_enum.py`) finds *all* equilibria, including of general-sum games.



**The pure-first hybrid.** On the deterministic game the "yes" branch is taken at every one of the 2380 states, so the LP is never called.

# RQ1 – Existence

| MOVE ORDER                   | STOCHASTIC | CONVERGED STAGE GAMES WITH NO PURE SADDLE | STATIONARY PURE EQ? |
|------------------------------|------------|-------------------------------------------|---------------------|
| deterministic<br>(exact A10) | no         | 0 / 2380                                  | yes (Claim B)       |
| coinflip tie-<br>break       | yes        | 0 / 2380                                  | yes (Claim B)       |
| random move<br>order         | yes        | 94 / 2380 (this config)                   | no                  |

IT IS THE MOVE ORDER, NOT THE RANDOMNESS

The deterministic and coin-flip value functions are *identical* ( $\max |V_{\text{det}} - V_{\text{coin}}| = 0$  at every  $\gamma$ ): optimal play never enters a losing contest, so the coin is never flipped. Only Littman's random move order — where a ball-steal depends on who resolves first *and* on both players' targets — creates matching-pennies subgames.

## RQ2 – Efficiency

RANDOM MOVE ORDER,  $\Gamma = 0.9$  – THE CASE WHERE THE LP IS ACTUALLY NEEDED

| SOLVER              | SWEEPS | V(KICKOFF) | MATRIX-GAME SOLVES | WALL CLOCK |
|---------------------|--------|------------|--------------------|------------|
| pure (lower bound)  | 18     | +0.000     | 0                  | <1 s       |
| hybrid <b>exact</b> | 108    | +0.150     | ~9 700             | ~14 s      |
| mixed <b>exact</b>  | 108    | +0.150     | ~3.6 M             | ~8 min     |

The hybrid solves an LP on only the ~4% of states lacking a pure saddle — ~25× fewer — and matches the all-LP value to 4e-16. **Mirror symmetry** (`run_symmetric`): the game is anti-symmetric under board-flip + player-swap and every state has a distinct mirror, so the 2380 states are 1190 pairs — solve one, reconstruct the other by  $V(\text{mirror}(s)) = -V(s)$ .

## Freeze-then-iterate, and its staleness

RANDOM MOVE ORDER, HYBRID SOLVER – SAME FIXED POINT,  $|\text{DIFF}| < 3\text{E-}9$

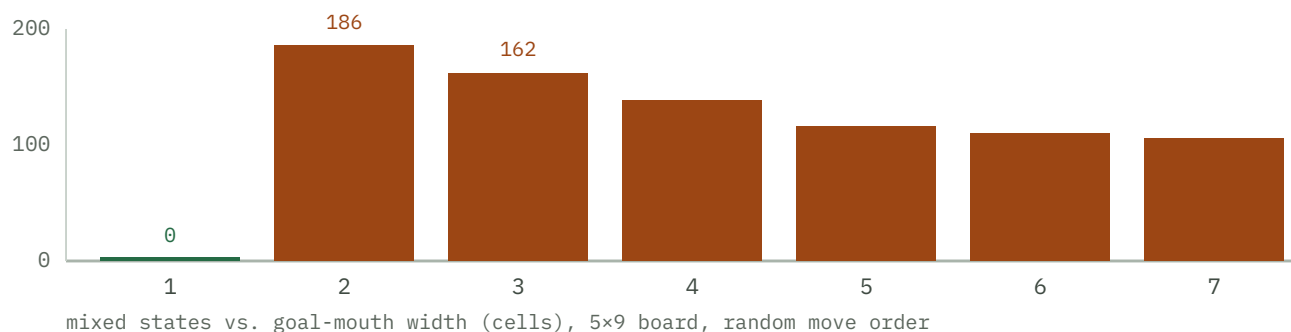
|                  | ROUNDS / SWEEPS | MATRIX-GAME SOLVES | WALL CLOCK |
|------------------|-----------------|--------------------|------------|
| value iteration  | 108             | 9 672              | 13.8 s     |
| policy iteration | 21              | 1 879              | 17.2 s     |

5× fewer LP solves — but the frozen strategies' distance to the current Nash stays *maximal* (a full pure-strategy flip) for **16 outer rounds**, then collapses to  $< 1\text{e-}8$ . The thrashing eats the saving; on the deterministic game it is strictly worse. Value iteration with the per-sweep Nash cache wins here.

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## RQ3 – Mechanism

A stage game needs mixed strategies only when the goal mouth is more than one cell wide.



**The gate.** One-cell goal → the carrier has one threat, the defender one response, best responses stay pure. Two or more → the defender must guess. Every height-3 board (width 3–11) has zero mixed states.

Geometry predicts the label (`scripts/geometry_model.py`): a depth-4 decision tree on carrier-frame features separates mixed from the rest with **precision 0.96, recall 0.91**. The dominant feature is `defender_can_intercept` — the defender within one move of the carrier's forward cell ( $P(\text{mixed}) = 0.44$  vs  $0.002$ ).

Beyond prediction, the 94 mixed states canonicalize under the board mirror to 47 pairs and cluster into **8 geometric templates** (`scripts/templates.py`, `docs/templates.md`).

## THE MIXED-STATE TEMPLATES – CARRIER-FRAME GEOMETRY, EQUILIBRIUM SUPPORT

| STATES | GEOMETRY                                               | SUPPORT | STRUCTURE             |
|--------|--------------------------------------------------------|---------|-----------------------|
| 24     | defender 1 ahead, carrier one row off the goal row     | 2×2     | matching pennies      |
| 24     | defender 1 ahead, same row, carrier on a goal row      | 2×2     | matching pennies      |
| 16     | defender 2 ahead, same row, carrier on a goal row      | 2×2     | matching pennies      |
| 4      | defender diagonally adjacent, carrier off the goal row | 2×2     | matching pennies      |
| 14     | defender 1–2 ahead, same row, on a goal row            | 2×1     | near-pure saddle      |
| 8      | defender 2 ahead, same row, on a goal row              | 3×2     | degenerate (row ties) |
| 4      | defender 2 ahead, carrier off the goal row             | 3×3     | wider mix             |

The four 2×2-support templates are all verified matching pennies — the carrier's best reply flips between the two defender columns and vice versa — covering 68 of 94 states: carrier {climb, advance} against defender {cover a lane, hold the forward cell}. All 94 share one value across every equilibrium; 64 have a unique one.

**Why the mix is unavoidable.** A mixed stage game needs all three of: a stochastic ( $\frac{1}{2}$ – $\frac{1}{2}$ ) resolution order; a goal mouth wider than one cell, so the carrier has two scoring lanes; and a defender close enough to contest the forward cell but unable to cover both lanes in one move. The carrier climbs or drops, the defender guesses which — matching pennies. Drop any one clause — a single goal cell, deterministic resolution, or the coin-flip tie-break — and every stage game has a pure saddle. The full argument and one stage matrix per template are in docs/templates.md; a formal proof for the single-cell case is open (docs/mechanism.md).

## RQ4 – Numerical robustness

### The value is three numbers

`value_bracket` returns what the row player guarantees ( $\min_k (pM)_k$ ), gets ( $p^T Mq$ ), and can reach ( $\max_i (Mq)_i$ ). The true value is bracketed, so the width certifies the LP error.

FINDING

With `scipy`'s HiGHS the three agree to  $1.1e-16$  per stage game,  $8.7e-10$  accumulated. The concern is real for older vertex/simplex solvers — a non-issue here.

### Discounting and tolerance

|                                                                     | DETERMINISTIC | RANDOM (7×5)                |
|---------------------------------------------------------------------|---------------|-----------------------------|
| mixed states, $\gamma = 0.5 \rightarrow 0.9 \rightarrow 0.995$      | 0 → 0 → 0     | 122 → 94 → 102              |
| classification split, <code>rel_tol</code> $1e-12 \rightarrow 1e-3$ | stable        | 604 / 1682 / 94<br>(stable) |
| classification, <code>rel_tol</code> $1e-2 / 1e-1$                  | —             | 80 / 0 mixed                |
| fixed 0.1-rounding flips a saddle                                   | 0             | 62                          |

Discounting mildly shifts which states mix; the classification is otherwise robust across nine orders of magnitude of tolerance, breaking only as the tolerance approaches the real minimax – maximin gaps. Fixed 0.1-rounding — proposed to force indifference — misclassifies 62 states, exactly the small-entry mixed region.



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## Limitations

- The A10 page redacts the ID-specific start position and goal rows; this repo uses the standard Littman geometry (configurable). Qualitative results hold across 40+ board configurations; the exact 3.9% is the 7×5 / 3-cell-goal case.
- Several collision sub-cases (carrier vs. stationary opponent, bump = steal, swap possession) are *interpreted* from the two stated A10 rules, not quoted. If course staff intended otherwise, Claim A must be re-measured. See docs/assumptions.md.
- **Claim A** (every converged stage game has a pure saddle) and **Claim B** (a stationary pure equilibrium exists) are established by enumeration. **Claim C** (the theoretical game necessarily has one, over all settings) is *not* — that needs a proof.
- random and coinflip are different game definitions, isolating the collision rule and the tie-break respectively — not the A10 game.

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## Secondary validation — the policy plays correctly

- Duality gap  $V0_{br}(s_0) + V1_{br}(s_0) < 1e-9$  for every move order — no opponent beats the game value.
- Nash vs. Nash reproduces the value: forced draws in the deterministic and coin-flip games; +0.157 empirical discounted return (vs. +0.150 computed) in the random game.
- A best response to one assumed opponent is fragile: the Part 2 policy has exploitability 0.43 — worse than random — which is the A10 competition's warning about non-equilibrium submissions.

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## Beyond zero-sum, and open questions

- **General-sum — extension point.** `markov_game.py` solves 2-player general-sum Markov games by enumerating *all* stage equilibria (`support_enum.py`) and selecting one — the notes' "largest sum of values", a no-op for zero-sum but decisive for Battle of the Sexes, where it avoids the mixed equilibrium whose value is below either pure one. Pure-first throughout. The soccer game is zero-sum, so this does not touch the main result.
- **Function approximation — partial.** A network fit to the stage matrices (`nash_dqn.py`,  $5 \rightarrow 96 \rightarrow 96 \rightarrow 16$  with biases, 500 epochs) trails the exact solver badly:

|                               | EXACT HYBRID NASH-Q | NEURAL NASH-Q    |
|-------------------------------|---------------------|------------------|
| wall clock                    | 9 sweeps, 0.45 s    | 500 epochs, 55 s |
| $\max  V - V_{\text{exact}} $ | 0                   | 0.54             |
| action agreement              | 100%                | 53%              |
| exploitability                | 0                   | 0.43             |

The network fits the  $\sim 1000$  zero-value states but misses the  $\gamma^k$  structure — about as exploitable as random play. Keep the exact solver as ground truth.

- **A proof (open).** Can "one-cell goal  $\Rightarrow$  pure everywhere" be turned into a theorem about the transition structure?

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soccer_nash — game.py · matrix_games.py · support_enum.py · nash_q.py · markov_game.py ·
numerics.py · geometry.py · symmetry.py · tree.py · best_response.py · exploit.py · mlp.py
· nash_dqn.py · opponents.py · shaping.py · simulate.py · experiment.py · render.py ·
templates.py
scripts — experiments.py · analyze.py · numerics.py · equilibria.py · geometry_model.py ·
templates.py · policy_iteration.py · selfplay.py · nash_dqn.py · render.py · a10_part1.py
· a10_part2.py · a10_competition.py
experiments — baseline.csv · gamma_sweep.csv · tolerance_sweep.csv · board_sweep.csv ·
goal_mouth_sweep.csv · one_cell_goal.csv
env: 7×5 grid, 2380 states, zero-sum  $\pm 1$ ,  $\gamma = 0.9$ 
```